



Optimization of Real-Time GPS Queuing Delays in City Transport

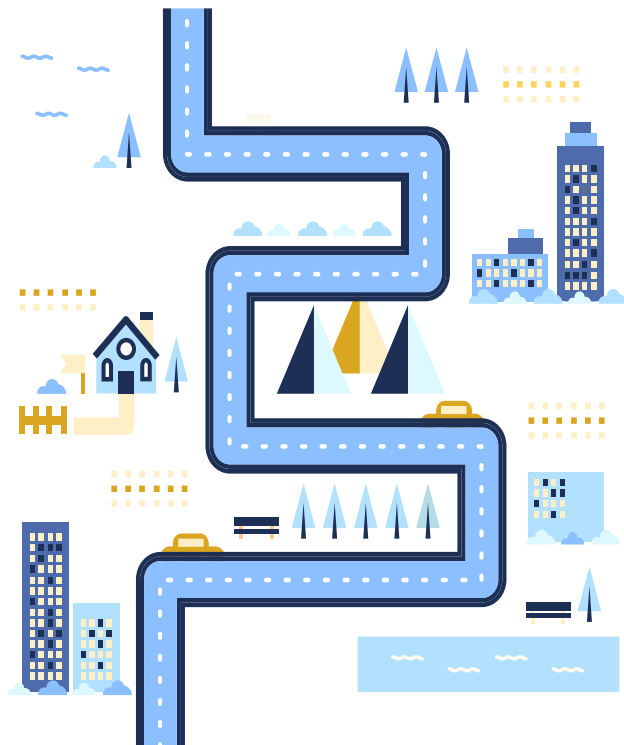
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OBJECTIVE

SCENARIO

Smart cities are relying on the accuracy of real-time bus-tracking

- Commuters depend on the mobile apps up-to-date ETA and bus locations
- Outdated information leads to frustrated passengers and inefficient transport use
- During peak hours there can be network congestion issues



CURRENT PROBLEM

- Simultaneous GPS updates from multiple buses during rush hour
- Network routers incurring queuing delay
- Results in :
 - Outdated ETA displays
 - Poor user experience within the app
 - Missed bus tracking events

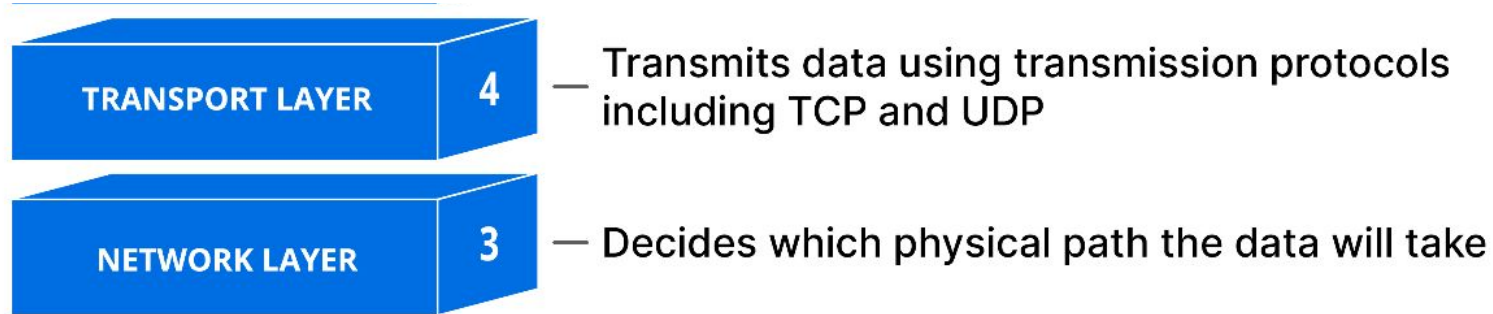
Network Layers Involved

Network Layer:

- Routing and packet forwarding decisions

Transport Layer:

- Reliable data delivery and congestion control



DESIGN QUESTIONS



ARCHITECTURE TYPE

What network is
best suited for
this design?



QUEUING STRATEGIES

What queuing
strategies are
going to be
tested?



TRAFFIC CONFIGURATION

How to mimic a
high traffic load
of buses?



PROOF OF DESIGN EFFICIENCY

What outcomes
prove the
efficiency of the
selected
design?

Queuing Strategy: FIFO (First In, First Out)

This queuing strategy is to be implemented at the router and is used as queue management of the incoming packets

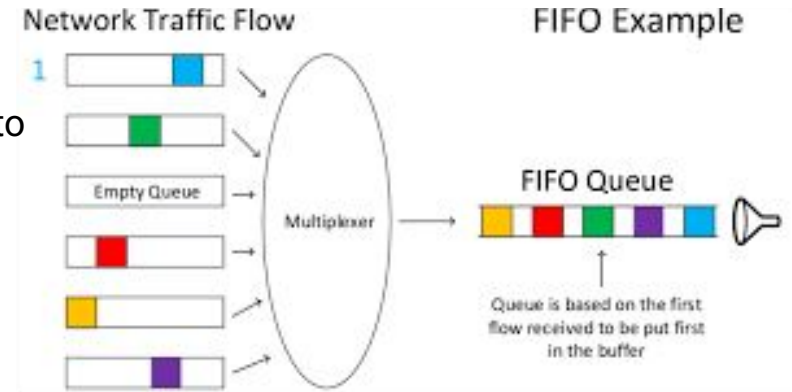
- Used as the baseline of this project

Configuration:

- Access "Router's Interface properties" > set "Queuing type" to **FIFO**

How FIFO functions:

- Treats all packets equally
- Queues in the order received
- no distinction between a GPS updates and a Large file download



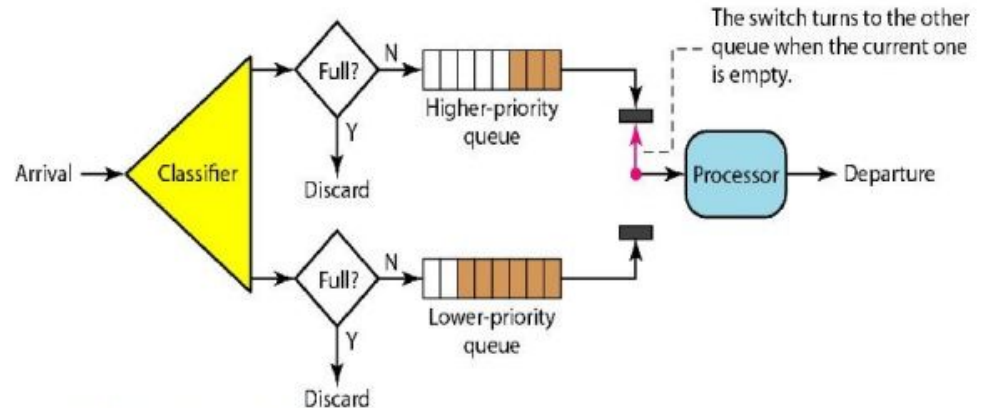
Queuing Strategy: Priority Queuing

Configuration:

- Access “Router’s Interface properties” > set “Queuing type” to **PRIORITY**

How Priority Queue Works

- Router creates distinct queues
 - RTPS tag (GPS) = “highest priority”
 - BE tag (congestion) = “lowest priority”
- Router always empties the high priority queue before sending anything from the low priority queue



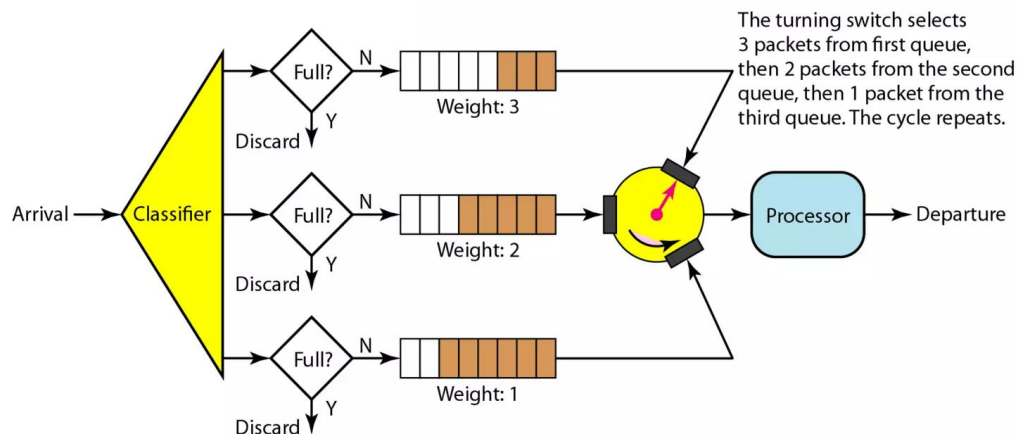
Queuing Strategy: Weight-Fair Queuing

Configuration:

- Access "Router's Interface properties" > set "Queuing type" to **WFQ**

How Priority Queue Works

- Router creates distinct queues
- assigns a weight to each one
- Attempts to share bandwidth fairly
- round-robin fashion



SCENARIO CONFIGURATIONS

Network Architecture

In this project we are using LTE (Long Term Evolution) Topology. LTE provides Wide Area Network coverage and high speed transfer capabilities suitable for city buses.

Net-Sim Module

LTE/LTE-Advanced was used to simulate this project. It features mobile nodes (UE) and mimicking WAN in a Smart City.

Mobile Nodes

10 UEs were used to behave like buses moving at 15 m/s. RANDOM_WAYPOINT was used to mobilize the buses. These buses are all assigned the RTPS tag to mark it as real-time delivery.



Architecture & Links

1. Why LTE instead of Wi-Fi?

Coverage: Wi-Fi (WLAN) has a range of only ~100 meters, requiring thousands of access points for a city route. LTE (WAN) covers kilometers with a single tower.

Mobility: LTE handles high-speed handovers (vehicles moving at >50 km/h) natively. Wi-Fi connections are unstable at high speeds.

2. Mobility Model Configuration

Model: Random Waypoint.

Velocity: 15 m/s (~54 km/h) to simulate average urban bus speeds.

Pause Time: 10 seconds to simulate bus stops for passenger boarding.

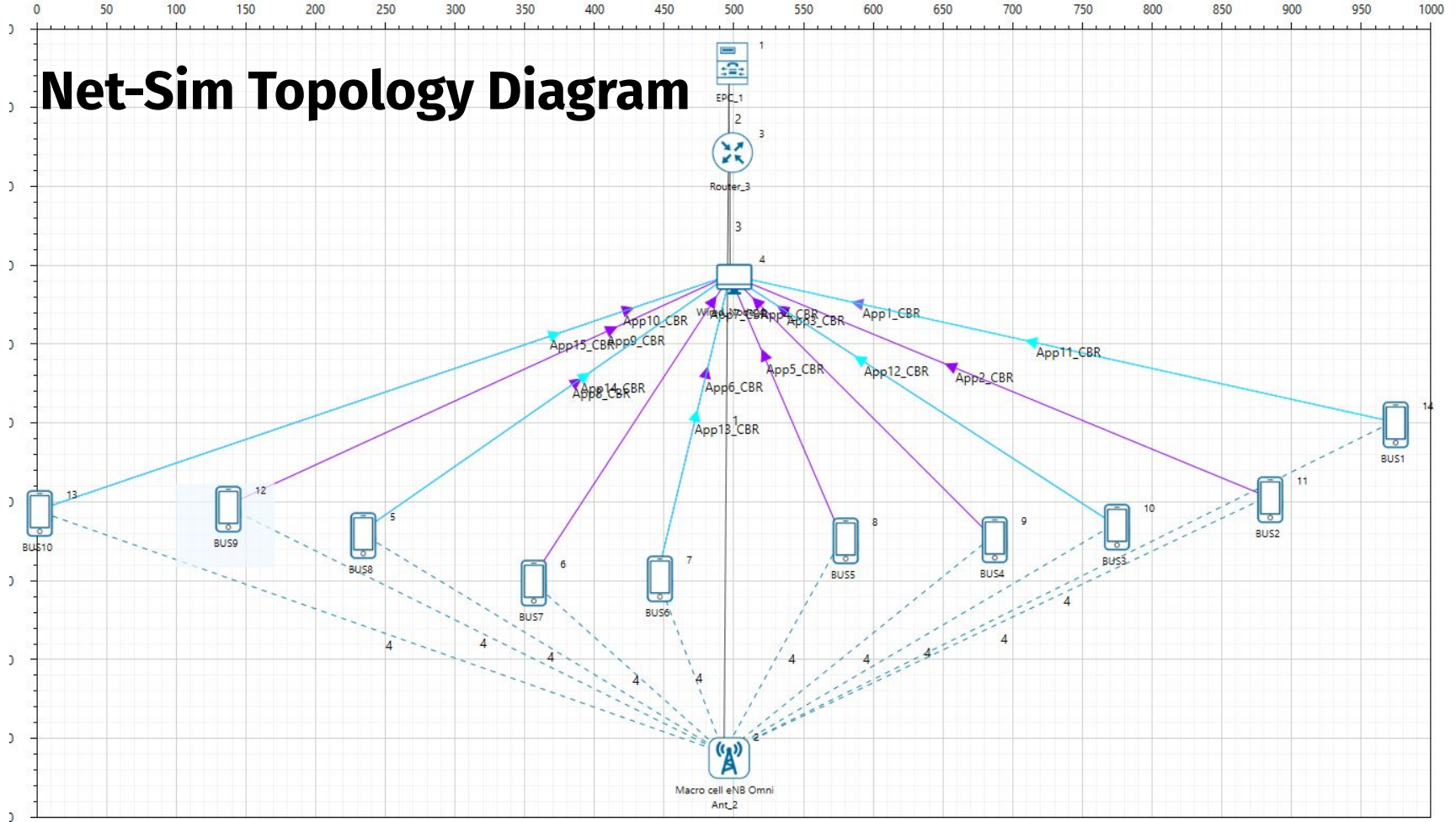
3. Link Configuration

Wireless Link: LTE Air Interface (Subject to interference and distance fading).

Wired Links (Backhaul): High-speed Gigabit links connecting the Tower ->EPC -> Router -> Server.

Design Logic: The wired links are over-provisioned (very fast) to ensure that the only delay in the system comes from the Router's queuing processing, not from cable bandwidth limits.

Net-Sim Topology Diagram



Network Components

1. Mobile Nodes (Blue Icons - UE)

Identity: These 10 nodes represent the City Buses.

Function: They are the source of all traffic (both the critical GPS data and the passenger congestion traffic).

Connection: They connect wirelessly to the central tower (eNodeB) via the LTE air interface.

2. eNodeB (Central Tower)

Identity: The LTE Base Station (Cell Tower).

Function: Provides Wide Area Network (WAN) coverage. It aggregates the wireless signals from all moving buses and forwards them to the wired core network.

3. Router_3 (Center Device)

Identity: The Network Bottleneck.

Function: This is the critical device for our experiment. It sits between the high-speed core network and the server.

Why here? We configure the queuing policies (FIFO, Priority, WFQ) on this router's interface to manage the bottleneck.

4. Wired_Node_4 (Right PC)

Identity: The Central Traffic Server.

Function: The destination for all GPS updates. We measure packet arrival times here to calculate latency.

Traffic Generation

Traffic Generation (The "Stress Test")

We configured a heterogeneous traffic mix to force competition for bandwidth:

- **Critical Traffic (GPS):**
 - Source: 10 Buses (100% of fleet).
 - Packet Size: Small (150 Bytes).
 - Priority Tag: RTPS (Real-Time Polling Service).
 - Interval: 1 second (Periodic updates).
- **Congestion Traffic (Background Noise):**
 - Source: 5 Buses (50% of fleet).
 - Packet Size: Large (1024 Bytes).
 - Priority Tag: BE (Best Effort).
 - Rate: High speed (~500 Kbps) to saturate the router queue.

Simulation Time: 100 seconds (Allows sufficient time for queue buffers to fill up).

Transport Protocol: UDP (Standard for real-time telemetry).

Baseline Simulation Setup

1. Configuration

Queuing Discipline: FIFO (First-In, First-Out).

Router Behavior: The router treats all incoming packets equally. It places them in a single buffer in the exact order they arrive.

Traffic Condition:

- 10 flows of small GPS packets (150 Bytes).

- 5 flows of massive Congestion packets (1024 Bytes).

2. The Mechanism of Failure

Head-of-Line Blocking: A small GPS packet arriving just milliseconds after a large file download packet must wait for the entire large packet to be processed.

Result: In a saturated network, the "line" is always long. Critical real-time data gets stuck behind non-critical bulk data.

Optimization Strategy

To solve the latency issue, we implemented Quality of Service (QoS) mechanisms at the Router interface:

Change 1: Priority Queuing (PQ)

Configuration: We enabled "Strict Priority."

Rule: The router was configured to identify the RTPS tag (GPS) as "High Priority."

Action: The High Priority queue is always emptied completely before the router sends a single bit from the Low Priority (Congestion) queue.

Change 2: Weighted Fair Queuing (WFQ)

Configuration: We assigned "Weights" to traffic classes.

Rule: Bandwidth is split dynamically.

Action: The GPS flow is given a guaranteed percentage of the link's capacity (e.g., 60%), ensuring it always has a path through the bottleneck regardless of congestion levels.

Justification for Optimization

1. Why Priority Queuing (PQ) Works

The "VIP Lane" Concept: It creates a dedicated express lane for GPS data. Since GPS packets are small (150 Bytes), they can be processed instantly without waiting for the large 1024-byte congestion packets to finish serialization.

Expected Outcome: Near-zero latency for GPS, at the risk of potentially "starving" the congestion traffic if GPS volume becomes too high.

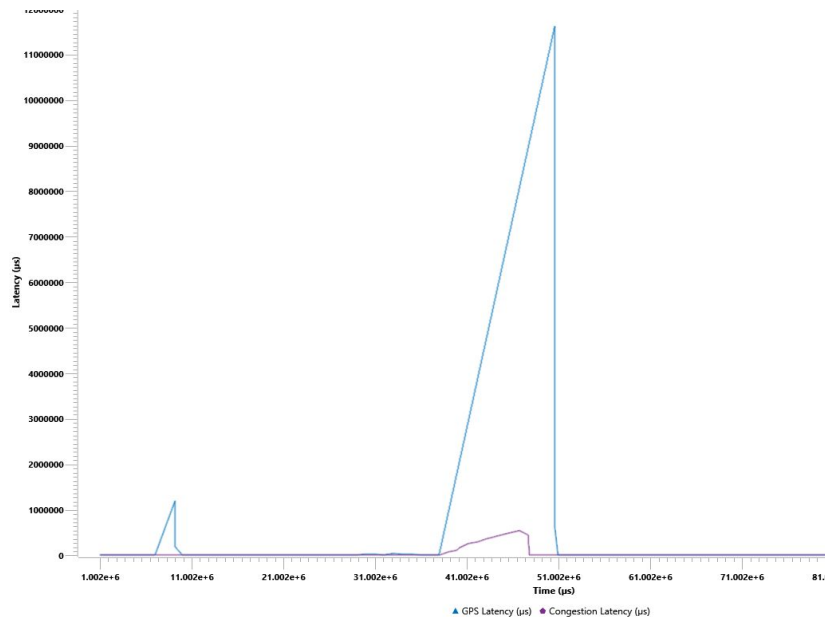
2. Why Weighted Fair Queuing (WFQ) Works

The "Fair Share" Concept: Unlike FIFO (which is unfair to small packets) and PQ (which can be unfair to low-priority packets), WFQ ensures every flow gets a turn.

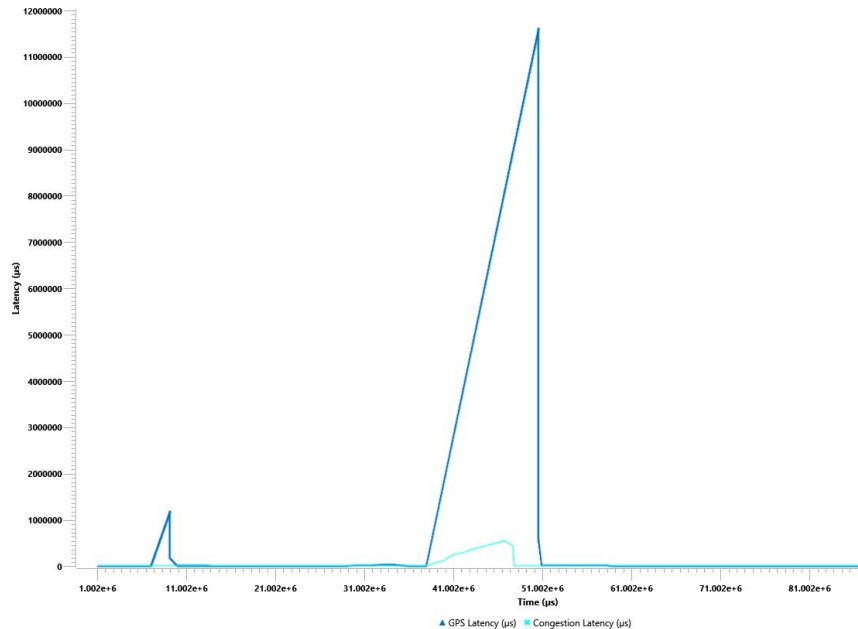
Efficiency: Because GPS packets are so small, they finish their "turn" in the WFQ round-robin cycle extremely fast, effectively achieving the low latency of PQ while maintaining network fairness.

Latency Results

Priority Queue



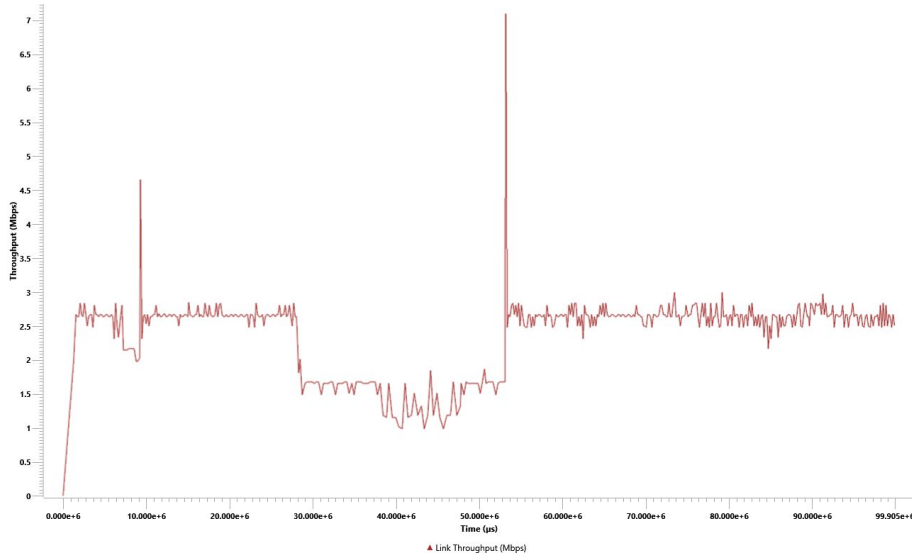
Weight-Fair Queuing



This is a comparison of Latency from a GPS packet (dark blue) and a Congestion packet (alternate color). These are both the same packets being compared

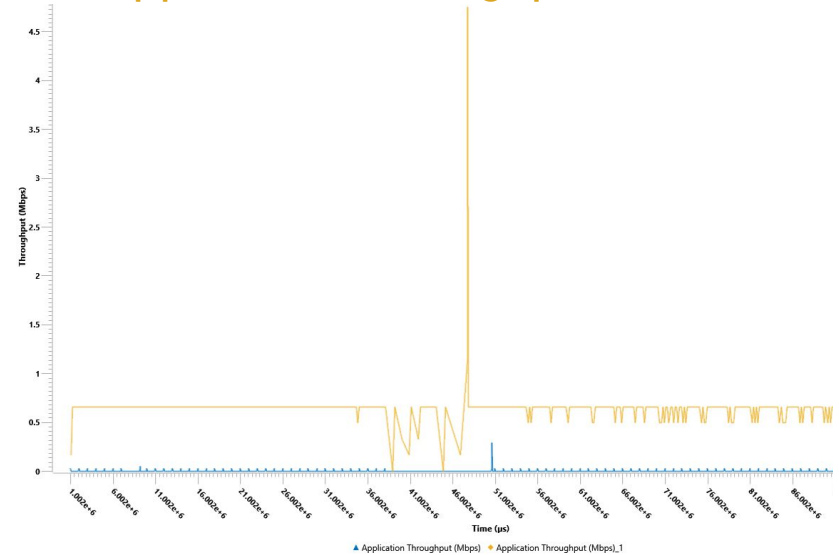
Throughput Results

Link Throughput



Throughput from router to the server.

Application Throughput



Throughput of the congestion packets (yellow) and the GPS small packets (blue)

GPS Latency Comparison (Average Delay)

Queuing Discipline	Average Delay (ms)	Condition	SLA Compliance
FIFO (Baseline)	760.0 ms	Severe Head-of-Line Blocking	Fails
Priority Queuing (PQ)	1.5 ms	Near Real-Time	Passes
Weighted Fair Queuing (WFQ)	1.5 ms	Near Real-Time	Passes

1. **FIFO Failure:** The baseline experiment established that standard FIFO queuing results in an unacceptable 760 ms delay. This is a critical system failure, as the data is outdated (stale) before it can be used for real-time tracking.
2. **QoS Success:** Implementing either Priority Queuing or WFQ immediately resolved the bottleneck, reducing the latency by 99.8% to just 1.5 ms.
3. **Key Insight:** This proves that the router, when instructed, can successfully isolate the small, high-priority GPS packets (RTPS) from the large congestion packets (BE).

Reliability and Congestion Impact

Packet Delivery Ratio (PDR) for both Critical and Congestion Traffic.

Queuing Discipline	GPS PDR	Congestion PDR	Key Finding
FIFO	100%	~70%	High delay, but no packet loss for critical packets.
Priority Queuing (PQ)	100%	~70%	The router prioritizes delivery <i>speed</i> without sacrificing GPS reliability.
Weighted Fair Queuing (WFQ)	100%	~70%	Reliable delivery maintained across all scenarios.

- 1. **GPS Reliability:** In all three experiments, the GPS traffic achieved **100% Packet Delivery Ratio (PDR)**. The problem was never packet loss; it was packet serialization delay.
- 2. **Congestion Confirmation:** The low PDR for the Congestion traffic (~70%) confirms that the network link was indeed saturated. The router was forced to manage an overflow situation.
- 3. **Conclusion:** The goal was met without requiring retransmissions or increasing network complexity, as the critical data was successfully protected from queuing drops.

Final Recommendation: PQ vs. WFQ

1. Priority Queuing (PQ):

- **Pros:** Achieves the fastest possible latency (1.5 ms). Simple to configure.
- **Cons (The Risk):** In high-traffic scenarios, PQ can lead to "starvation," meaning low-priority services (like user downloads) may never be served.

2. Weighted Fair Queuing (WFQ):

- **Pros:** Achieves the same low latency (1.5 ms) for small packets. *Crucially*, it guarantees a minimum share of bandwidth for the low-priority traffic, ensuring fairness.
- **Cons:** Slightly more complex in configuration (if weights were manually editable).

We recommend **Weighted Fair Queuing (WFQ)** for the smart city network. It provides **real-time performance** (1.5 ms) necessary for GPS tracking while maintaining a **balanced distribution of resources**, which is vital for providing fair service to all citizens using the public Wi-Fi on the buses.

Performance Calculation (Delay Reduction)

This calculation quantifies the effectiveness of our design by measuring the percentage reduction in latency from the baseline.

$$\text{Delay Reduction (DR)} = \frac{\text{Delay}_{\text{FIFO}} - \text{Delay}_{\text{QoS}}}{\text{Delay}_{\text{FIFO}}} \times 100$$

Calculation:

$$\text{DR} = [(760 \text{ ms} - 1.5 \text{ ms}) / 760 \text{ ms}] \times 100 = 99.8\%$$

Result:

The implementation of QoS reduced GPS traffic latency by 99.8%, eliminating the Head-of-Line Blocking issue.

Network Saturation Calculation (Packet Drop %)

This calculation proves that the network was successfully congested, validating the stress test conditions.

Packet Drop % = (Packets Generated - Packets Received)/
Packets Generated × 100

- Generated: 6,043 packets
- Received: 4,383 packets

Packet Drop % = $(6043 - 4383) / 6043 \times 100 = 27.47\%$

The congestion traffic (BE) experienced a 27.47% packet drop rate, confirming that the network link was saturated and forced the router to prioritize.



Conclusion (Design Effectiveness)



The results provide complete support for the effectiveness of our design:

1. **Justification for QoS:** The **760 ms** delay in the FIFO experiment validates the critical need for traffic differentiation.
2. **Effectiveness:** The reduction in latency to **1.5 ms** with PQ and WFQ confirms that tagging the GPS packets as **RTPS** successfully protected the data stream from congestion.
3. **Final Recommendation:** **Weighted Fair Queuing (WFQ)** is the optimal choice, as it achieves the necessary low latency while ensuring network fairness for non-critical user traffic.

Constraints & Simplifications

1. Simulation Constraints (NetSim Limitations)

- **Pause Time Fix:** The software imposed a hard maximum of **10 seconds** for the "Pause Time" mobility parameter. This artificially limits the duration of real-world bus stops.
- **WFQ Weighting:** Manual adjustment of Weighted Fair Queuing (WFQ) class weights was unavailable in the router interface. We relied on the standard NetSim WFQ algorithm, which performed well but limits our ability to fine-tune the fairness ratio (e.g., 90/10 split).
- **LTE Model:** The current LTE model does not account for complex interference patterns (inter-cell interference) or real-world signal fading specific to urban canyon environments, simplifying the wireless channel.

2. Mobility Simplification

- **Model Used:** Random Waypoint.
- **Constraint:** This model uses random destinations and paths. In reality, buses follow **fixed routes**. This simplification may slightly underestimate the impact of cell edge effects on specific route segments.

Future Work & Project Extensions

1. Architecture & Testing

- **5G NR Integration:** Test the scenario using **5G New Radio**. Analyze if 5G's extremely low latency eliminates the need for Layer 3 QoS, or if traffic slicing is required.
- **Handoff Analysis:** Implement fixed bus routes and multiple towers to measure specific delay and loss experienced during **cell-to-cell handover**.

2. Scalability & Load

- **Increased Fleet Size:** Scale the number of UEs (buses) to 50 or 100 to determine the congestion limit before the high-priority queue fails.
- **Dynamic Load:** Implement time-varying traffic where congestion only peaks during defined "rush hour" intervals for a more realistic test.

3. Limitations

- The simulation used a simplified mobility model (Random Waypoint) instead of fixed city routes.
- Manual WFQ weight adjustment was limited.

Final Conclusion

Title: Conclusion & Final Validation

- **Problem Solved:** Implemented QoS successfully addressed the primary problem of **Head-of-Line Blocking** caused by FIFO queuing.
- **Quantitative Success:** Latency for critical GPS data was reduced by **99.8%** (from 760 ms to 1.5 ms), meeting the real-time SLA requirement.
- **Recommendation:** The optimal solution for the urban transport network is **Weighted Fair Queuing (WFQ)**.
 - It provides the necessary **RTPS** prioritization for safety and accuracy.
 - It maintains **fairness** for non-critical user traffic, improving overall network health compared to strict Priority Queuing.
- **Validation:** The project successfully demonstrated that Layer 3 QoS mechanisms are essential for protecting critical data streams in congested wireless networks.

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Thanks for listening to us!